Incorporating spatial information into non-negative matrix factorization

Inkorporering af spatial information i ikke-negativ matrixfaktorisering

Bachelor project in data science

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Abstract

Non-negative matrix factorization (NMF) is a method used for factorising a non-negative data matrix into two non-negative factor matrices. These factors allow meaningfully analysis of the underlining structures defining the data matrix. NMF is especially popular when studying genetic data since the non-negative restraint of the factors proposes an enhanced interpretability. When analysing genetic data it is expected that the spatial location of the cells affect their genetic expressions. The introduction of spatial transcriptomics (ST) into the world of genetic data has just put an increased focus on the specific spatial location of the individual cells when carrying out analysis of their genetic expressions. The standard NMF method however does not include any spatial information when estimating its factors and are thereby omitting crucial information when used to analyse genetic data.

We therefore propose three modified NMF methods which includes the spatial location of the cells when estimating the factors. The first method introduces a neighbourhood average over one of the factor matrices, the second method uses the structure of a Gaussian Markov Random Field (GMRF) to introduce a neighbourhood correlation and the last model is a simplified version of the second model. We further propose a way to estimate the parameters of a GMRF and thereby allowing us to estimate the spatiality of the factors estimated by the different NMF methods.

We evaluated all the proposed methods and the standard NMF method on a subset of the ST Slide-seqV2 mouse hippocampus dataset. The evaluation show that the first method produces a result with higher regard to the spatial structure of the data when compared to the result of the standard NMF. However the two methods based on the structure of GMRFs produces results that are unsatisfactory. We further show that the estimation of the spatiality of the factors estimated by the different NMF methods yields a satisfcatory result and correspond with the visual evaluation of the spatiality.

Finally we discuses the different ways that the unsatisfactory results might be improved through future development.

Resumé

Ikke-negativ matrix faktorisering (NMF) er en metod, der bruges til at faktorisere en ikke-negativ data matrix til to ikke-negative faktormatricer. Disse faktormatricer gør det muligt at udføre en meningsfuld analyse af de underliggende strukturer som ligger til grunde for datamatricen. NMF bruges i særdeleshed til at udføre analyser af genetisk data, da de ikke-negative antagelser om faktormatricerne bidrager til en øget fortolkelighed. I analyserer af genetisk data forventes det, at den spatiale lokation af cellerne påvirker deres genudtryk. Introduktionen af spatial transkriptomik (ST) til verden af genetisk data har blot forstærket fokuset på de specifikke spatialle lokationer af de individuelle celler, når der udføres en analyse af cellernes genudtryk. Den standard NMF metode inkluderer dog ingen spatial information når den estimerer faktormatricerne og udelukker dermed afgørende information, når den bruges til at analysere genetisk data.

Vi foreslår derfor tre modificerede NMF metoder, som inkluderer den spatialle lokation af cellerne når de estimerer faktormatricerne. Den første metode introducerer et nabogennemsnit over en af faktormatricerne, den anden metode bruger strukturen fra et Gaussian Markov Random Field (GMRF) til at introducere en nabokorrelation og den sidste metode er en simplificeret udgave af den anden metode. Vi foreslår ydereligere en metode til at estimerer parameterne i en GMRF og dermed gør det muligt for os at estimere spatialiteten af faktormatricerne estimeret af de forskellige NMF metoder.

Vi evaluerede alle de foreslåede metoder og standard NMF metoden på et subset af ST Slide-seqV2 mouse hippocampus datasættet. Evalueringen viste at den første metode producerer et resultat, der tager større hensyn til den spatialle struktur af data sammenlignet med resultatet fra standard NMF. Evalueringen viste dog yderelige, at de to metoder baseret på GMRF producerer et utilfredstillende resultat. Det viser ydereligere at estimationen af spatialiteten af de forskellige faktormatricer estimeret af de forskellige NMF metoder giver tilfredsstillende resultater og at disse stemmer overens med den visuelle vurdering af spatialiteten.

Til sidst diskuterer vi de forskellige måder, hvormed det utilfredsstillende resultat potentielt kan forbedres igennem fremtidig udvikling.

General Abbreviations

We use the following abbreviations throughout the entire paper

- NMF: Non-negative Matrix Factorization
- GMRF: Gaussian Markov Random Field
- ST: Spatial Transcriptomics

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0 Structure of the project

In the following section we describe the structure of the project.
In section 1 we introduce NMF and motivate the inclusion of spatial information of cells when analysing their gene expression using NMF. We further describe the data used to evaluate the methods proposed in the project in this section. In section 2.1 we introduce the theory behind the standard NMF. We then introduce the first of the three modified NMF models, Simple Spatial NMF, in section 2.2. Then we introduce the concept of GMRF in section 2.3 and explain how it can be used to estimate spatial correlation through two examples, autoregression and p-neighbourhood GMRF, in section 2.3.1 and 2.3.2, respectively. Then we introduce the theory behind the second modified NMF model in section 2.4 which incorporates the concept of p-neighbourhood GMRF into NMF to capture spatial information. The theory behind the last modified NMF model is introduced in section 2.4.1 which is a simplified version of the second model. Then the first and second modified NMF model are compared in section 2.5.
The models are then evaluated on the described data and their results are presented in section 3. These results are discussed in section 4 together with the future perspective of the models. In section 5 we conclude on the project. Further details of the models are shown in section 6 together with the corresponding algorithms. We have implemented all shown algorithms in Python, which are found in [1].

We have generally both taken part of the work behind all sections.
Christoffer authored section 2.2, 2.3, 6.2, 6.3 and 6.5
Bjarke authored section 2.1, 2.4, 5, 6.1 and 6.5
Sections 1, 2.5, 3 and 4 were authored together.

1 Introduction

Non-negative matrix factorization (NMF) is a widely used tool to analyse high-dimensional data, which provides an interpretable factorised representation of the data, and enables meaningful analysis of these factors [2]. In simplicity NMF is a unsupervised dimensionality reduction method which decomposes a data matrix V into two factor matrices W and H. There exist a multitude of other unsupervised dimensionality reduction methods which decomposes a data matrix into factor matrices. This is methods such as Principle Component Analysis, Linear Discriminant Analysis and Independent Component Analysis [12]. All these methods have something in common: none of them imposes any constraints on the sign of the elements in the factor matrices. In other words, they allow for the elements to have a negative sign. It is, however, not always beneficial to have negative signs in these factors since negative values in some real world cases are meaningless. This is for example the case when dealing with image or gene data analysis tasks [12]. However, NMF has been developed to handle exactly such data analysis tasks. Here NMF introduces a nonnegativity constraint on the factor matrices meaning the elements of the factor matrices W and H are not allowed to have negative signs. The nonnegativity constraint results in an enhanced interpretability when dealing with cases where negative values are meaningless e.g. a positive value can be interpreted as indicating the presence of an event and a zero the absence of an event [12].

More specifically, given a non-negative data matrix V, the goal of NMF is to find non-negative matrix factors W and H, referred to as the weight matrix and the signature matrix respectively, such that V is approximated by WH:

$$V \approx WH.$$

The matrix V has the dimensions $N \times M$, where each row is a M-dimensional data vector and N is the number of data vectors in the data set, hence V_{nm} represents the expression of type m in data vector n for $n = 1, ..., N,\ m = 1, ..., M$. The factor matrices W and H are, respectively, $N \times K$ and $K \times M$ matrices. The k'th row of H, H_k for $k = 1, \ldots, K$, is called a signature and is a probability vector over the possible M types present in the data vectors, i.e. H_{km} is the probability of type m occurring in signature k. The k'th column of W, $W_{(k)}$, is called a weight and represents the exposure of signature k in the data vectors i.e. W_{nk} indicates the exposure of signature k in data vector n.

One of the biggest use cases of NMF is to model biological processes. Here NMF has for example been used to model which genes are expressed in certain cells [10]. When modelling such biological processes the underlying spatial resolutions of the cells are crucial [11]. The introduction of spatial transcriptomics (ST) has made it easier and more accessible to retrieve data with a high spatial resolution when working with gene expressions in biological tissue's cells [4]. For each cell ST data consists of discrete counts of thousands of features, often genes. It further stores the spatial coordinates of each cell. Since the ST data simply is high-dimensional count data with spatial coordinates for each cell, it is possible to evaluate the data using NMF [10]. However, NMF ignores the spatial information available for each cell in the ST data and thereby fails to include crucial information when modelling the biological processes underlying

the data [10]. In order to use this crucial information it would be beneficial to develop a modification of NMF which incorporates the spatial information from ST data into the model.

The goal of this project is to develop a model which incorporates the spatial structure of such data into the NMF model. In order to achieve this goal we propose three models: The first model introduces a simple neighbourhood average of the weights based on the cells spatial structure in the estimation of W. The second model imposes an assumptions in regards to the distribution of W. Here it is assumed that each of the columns of W are a Gaussian Markov Random Field (GMRF) wrt. a graph, G, which is constructed based on the spatial structure of the cells. This assumption is then included in the estimation of W. Using GMRF allows for modelling the spatial correlation of the cells in a natural way through conditional dependence. In this model the parameters of the GMRF are estimated alongside the factors. The last model is a simplified version of the second model where it is assumed that all columns of W follows the same given GMRF.

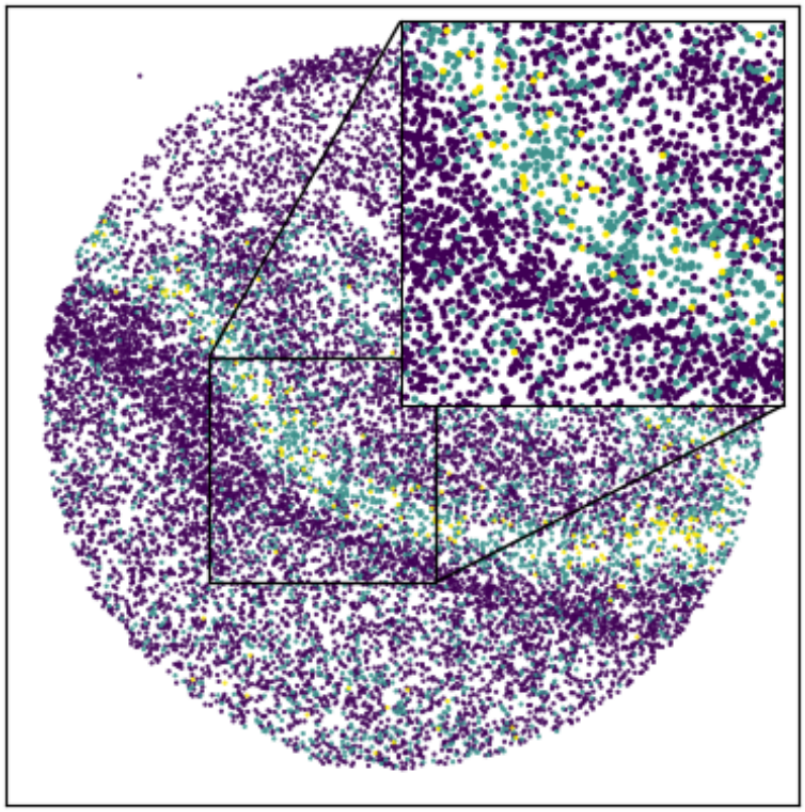

Figure 1: The spatial coordinates of the cells from the ST Slide-seqV2 dataset plottet for the gene Hpca. Each dot represents a cell. The square indicates the subset used for evaluation.

To evaluate the different models in this project, the ST Slide-seqV2 mouse hippocampus dataset found in [9] is used. This dataset consists of 36,536 cells from a 2-dimensional slice of hippocampus tissue from a mouse. The cells are represented as data vectors over the counts of the genes expressed in the corresponding cell. The data includes counts of about 4,000 genes for each cell. Since it is a ST dataset each cell further has a unique location in 2-dimensional space, which is represented by coordinates. In figure 1 the 2-dimensional coordinates of each cell is illustrated in the circle with the intensity of each

coordinate indicating the count of gene Hpca in the corresponding cell.
In our project only the cells that express more than 10 genes and the genes that are expressed more than 200 times are used in the analysis. This filtering allows the models to focus on only the most relevant data. To simplify the process, we further only evaluate the methods on a subset of the cells. This subset was chosen such that it represents a region with high spatial correlation for many of the genes in the dataset [10]. The chosen subset is marked by the square in figure 1. This final dataset has 2,370 cells and 459 genes, which results in a data matrix V with dimensions $2,370 \times 459$. This mean that for each cell, the dataset contains a count of each of the 459 possible genes. Additionally, each of the cells has corresponding spatial information, which is simply coordinates in 2-dimensional space.

2 Theoretical Setup

2.1 Standard NMF

In this section we will go through the process of estimating W and H in the standard NMF model. As described in section 1, the goal of NMF is to find non-negative matrix factors W and H of non-negative data matrix V, such that $V \approx WH$.

Since entries of the data matrix, in our case, represents counts of a given gene in a given cell, it is assumed that these are independent and follow a possion distribution:

$$V_{nm} \sim \text{Pois}((WH)_{nm}), \quad n = 1, ..., N, \quad m = 1, ..., M.$$

This is also beneficially since the negative log-likelihood of V is equal to the generalized Kullback-Leibler divergence between V and WH, denoted $D_{GKL}(V||WH)$, up to an additive constant. The log-likelihood is given by:

$$\begin{aligned}\ell(W, H|V) &= \underbrace{-\sum_n \sum_m \left[V_{nm} \log\left(\frac{V_{nm}}{WH_{nm}}\right) - V_{nm} + WH_{nm} \right]}_{D_{GKL}(V||WH)} + \underbrace{\sum_n \sum_m \left(V_{nm} \log(V_{nm}) - V_{nm}\right) - \log\left(V_{nm}!\right)}_{C} \\ &= -D_{GKL}(V||WH) + C.\end{aligned}$$

Finding W and H that maximizes the log-likelihood is thereby equivalent to finding W and H that minimizes $D_{GKL}(V||WH)$.
Given the high complexity of the structure of $D_{GKL}(V||WH)$, the minimum is found using numerical optimization methods. In this project, gradient descent is used. Since $D_{GKL}(V||WH)$ is not convex in both W and H together, it is unrealistic to find a true global minimum using this method [3]. However, a local minimum can be found. To find the update rules to be used in the gradient descent process we differentiate $D_{GKL}(V||WH)$ wrt. W_{nk} and H_{km}. As shown in section 6.1 this yields the following update

rules:

$$
\begin{aligned}
H_{km}^{i+1} &= H_{km}^{i} + \gamma_{km}^{i} \left[\sum_{n} W_{nk}^{i} \frac{V_{nm}}{(W^i H^i)_{nm}} - W_{nk}^{i} \right], \\
W_{nk}^{i+1} &= W_{nk}^{i} + \gamma_{nk}^{i} \left[\sum_{m} H_{km}^{i} \frac{V_{nm}}{(W^i H^i)_{nm}} - H_{km}^{i} \right],
\end{aligned}
\tag{1}
$$

where γ_{km}^{i} and γ_{nk}^{i} are learning rates. We transformed these into multiplicative updates by using specific learning rates for H and W as in [3]. These learning rates for H and W are given by:

$$
\gamma_{km}^{i} = \frac{H_{km}^{i}}{\sum_n W_{nk}^{i}}, \qquad \gamma_{nk}^{i} = \frac{W_{nk}^{i}}{\sum_m H_{km}}.
$$

This yields the following multiplicative update rules:

$$
\begin{aligned}
H_{km}^{i+1} &= H_{km}^{i} \frac{\sum_n W_{nk}^{i} \frac{V_{nm}}{(W^i H^i)_{nm}}}{\sum_{n'} W_{n'k}^{i}} = \left(H^i \otimes \frac{(W^i)^T \frac{V}{W^i H^i}}{(W^i)^T \mathbb{1}^{n \times m}} \right)_{km}, \\
W_{nk}^{i+1} &= W_{nk}^{i} \frac{\sum_m H_{km}^{i} \frac{V_{nm}}{(W^i H^i)_{nm}}}{\sum_{m'} H_{km'}} = \left(W^i \otimes \frac{\frac{V}{W^i H^i} (H^i)^T}{\mathbb{1}^{n \times m} (H^i)^T} \right)_{nk},
\end{aligned}
\tag{2}
$$

where $\otimes$ denotes element-wise multiplication and division between matrices denotes element-wise division. This notation is used throughout the rest of the paper.

Using multiplicative updates removes the learning rates as hyperparameters that has to be chosen before optimizing, and it ensures that, if the initialization is non-negative, the result are non-negative. Furthermore, the multiplicative updates have in this project empirically proven to converge faster than updates with other learning rates. This is however not guaranteed in general.
During each iteration of the gradient descent process the rows in H is normalized to sum to one. This is done to ensure two things: Firstly, it is done to incorporate the assumption of the rows in H being probability distributions, which the normalization makes it possible to interpret these as. Secondly, it ensures that the found factor matrices are unique in terms of scaling. This is not given without the normalization since W and H can be scaled: $WH = (WD)(D^{-1}H)$, where D is a diagonal matrix.
The process of estimating W and H through standard NMF is shown in algorithm 1, section 6.1.

2.2 Simple Spatial NMF

As mentioned in the introduction, the inclusion of spatial information can be crucial when trying to understand the molecular dynamics underlying a biological process [11]. The standard NMF model, as described in section 2.1, does not account for the spatial information of the cells and is thereby omitting crucial information. In order to improve the standard NMF model it would therefore be beneficial to implement the spatial information into the model in some manner. This could be done through a p-neighbourhood framework. For a cell j to be in the p-neighbourhood of cell i, cell j needs to be one of

the p closest cells to i and have i as one of its own p closest cells and $i \neq j$. This is denoted as $i \sim j$. An example of p-neighbourhoods with $p = 2$ for some 2-dimensional data is shown in figure 2.

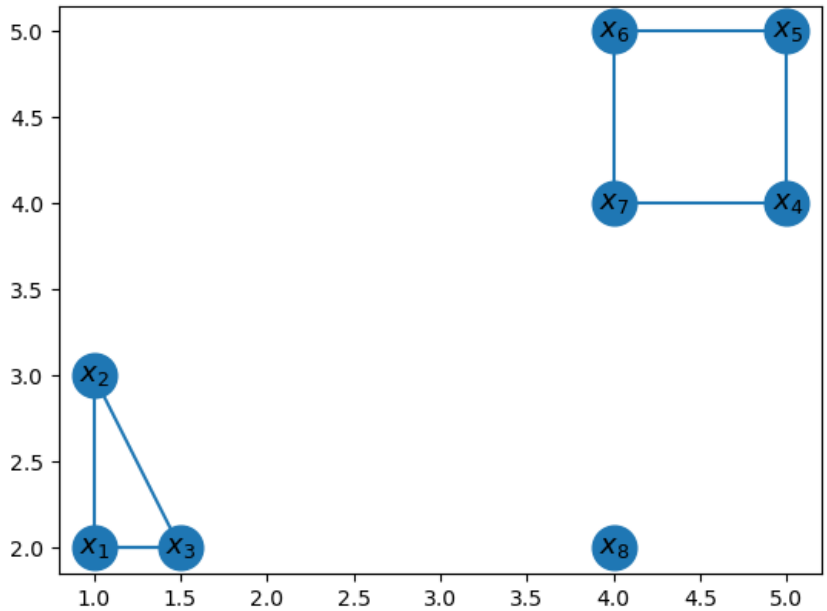

Figure 2: p-neighbourhoods with $p = 2$ for 2-dimensional data where an edge between x_i and x_j indicates that they are in each others neighbourhood.

Here we propose a simple method to incorporate the spatial information into the standard NMF using the p-neighbourhood framework. This new method still implements the multiplicative updates in (2). However, now in each iteration each weight is re-estimated using a p-neighbourhood average. The p-neighbourhood average of the k'th weight for cell n, W_{nk}, is calculated as the average of the entries $W_{n'k}$ where $n' \sim n$ and W_{nk}. The extra step is defined as:

$$W_{nk}^{i+1} = \frac{W_{nk}^{i+1} + \sum_{n' \sim n} W_{n'k}^{i+1}}{1 + \sum_{n'} \mathbb{1}_{n' \sim n}}, \tag{3}$$

where $\mathbb{1}_{n' \sim n} = \begin{cases} 1 & \text{if } n' \sim n \\ 0 & \text{else} \end{cases}$.

The process of estimating W and H through Simple Spatial NMF is shown in algorithm 2, section 6.2.

2.3 Gaussian Markov Random Field

As mentioned in section 1, GMRF allows for modelling of the spatial correlation between the cells. In this section we will introduce the theory behind a general GMRF and show how it can be used to model and estimate spatial correlation through two examples.

In general a random vector $x = (x_1, \ldots, x_n)^T$ is a GMRF wrt. to a graph $G = \{V, E\}$ with mean μ and precision matrix $Q > 0$, if its density has the form [7, p. 21-22]:

$$\pi(x) = (2\pi)^{-n/2}|Q|^{1/2}e^{-\frac{1}{2}(x-\mu)^T Q(x-\mu)},$$

and

$$Q_{ij} \neq 0 \iff \{i, j\} \in E \quad \forall i \neq j.$$

This is denoted as:

$$x \sim \text{GMRF}(\mu, Q).$$

This is equivalent to x being multivariate normal distributed with mean μ and covariance matrix $\Sigma = Q^{-1}$.

The entry Q_{ij} tells us about the conditional dependence between x_i and x_j [7, p. 21]:

$$x_i \perp x_j | x_{-\{i,j\}} \iff Q_{ij} = 0, \tag{4}$$

for $i \neq j$. We can thereby conclude that there only exists and edge (i, j) in E if x_i and x_j are conditional depended.

The conditional correlation between x_i and x_j can be formulated through the entries of Q as [7, p. 22]:

$$Corr(x_i, x_j | x_{-\{i,j\}}) = -\frac{Q_{ij}}{\sqrt{Q_{ii}Q_{jj}}}, \quad i, j = 1, \ldots n, \quad i \neq j. \tag{5}$$

In the following two sections we will go through examples of how we can use this structure of a GMRF to estimate the spatial correlation of x by imposing some further assumptions. The first example is a simple example to introduce the concept of estimating spatial correlation through a GMRF and the second example is the framework we use to estimate how spatial an estimated weight is and to modify the standard NMF model to incorporate spatial information.

2.3.1 Autoregression

To explain how the spatial correlation of the observations of a GMRF can be estimated we introduce a simple example of a GMRF namely an autoregressive process of order 1 with normal errors with mean zero and variance $\frac{1}{(1-\phi^2)}$. Here we assume the specific variance since it yields a nice structure of the corresponding precision matrix. Assuming $x = (x_1, \ldots, x_n)^T$ follows such an autoregressive process we can express the process as following

$$x_t = \phi x_{t-1} + \epsilon_t, \quad \epsilon_t \overset{iid}{\sim} N(0, \frac{1}{(1-\phi^2)}), \quad |\phi| < 1, \quad t = 1, \ldots, n. \tag{6}$$

It is seen that the observation x_t in (6) only depends on the previous observation x_{t-1}, a defining property of a Markov process. [7, p. 1]. It is further seen that ϕ determines how depended x_t is on the previous observation. In the top row of figure 3 three examples of the describe autoregerssion with different ϕ are shown. Here it is clear that when ϕ is larger then each observations is more dependent on the previous observation.

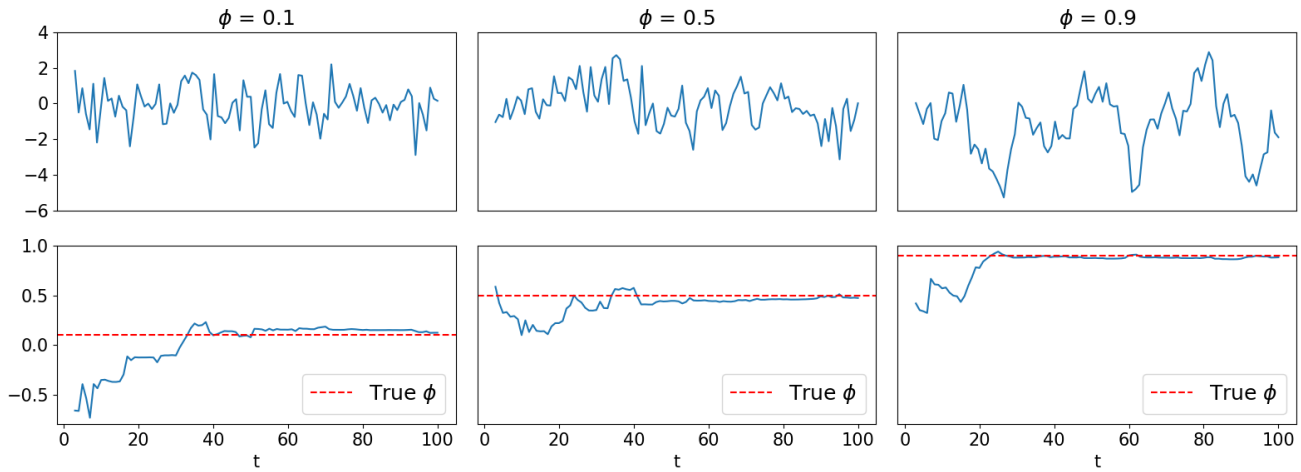

Figure 3: Simulation from the autoregressive process with different ϕ and $n = 100$ (top row) and the MLE of ϕ using the $t = 1, ..., 100$ first observations (bottom row)

In the above described case of autoregression there is no specific spatial information connected to x but instead the observations x_t are assumed to belong to different timestamps t. This concept of time can be used to create p-neighbourhoods like the concept of space as described in section 2.2. This is illustrated in figure 4.

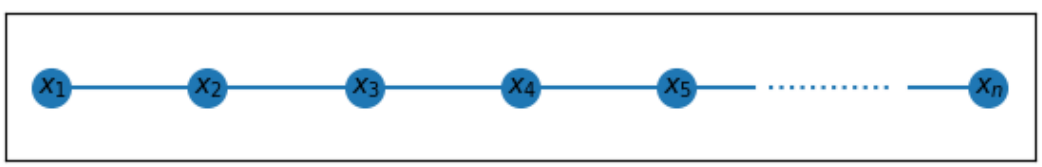

Figure 4: Graph of autoregression where there only exist an edge between two observations if they are p-neighbours in time with $p = 2$ and thereby conditionally dependent.

An essential assumption regarding the autoregressive process is the assumption about conditional independence. This assumption is not clearly stated in the expression above but can be shown if we write the process in its directed conditional form

$$x_t | x_1, \ldots, x_{t-1} \sim N\left(\phi x_{t-1}, \frac{1}{(1-\phi^2)}\right),$$

for $t = 2, \ldots, n$ [7, p. 1]. Here we see that the directed conditional only depends on the previous observation.

Given the above assumptions the joint density of x becomes

$$\pi(x) = \pi(x_1)\pi(x_2|x_1)\dots\pi(x_n|x_{n-1}) = \frac{1}{(2\pi)^{n/2}}|Q|^{1/2}e^{-\frac{1}{2}x^TQx}, \tag{7}$$

where Q is the precision matrix and is given by the following tridiagonal matrix

$$Q = \begin{bmatrix} 1 & -\phi & & & \\ -\phi & 1+\phi^2 & -\phi & & \\ & \ddots & \ddots & \ddots & \\ & & -\phi & 1+\phi^2 & -\phi \\ & & & -\phi & 1 \end{bmatrix}$$

We know from (4) that if $Q_{ij} = 0$ for $i \neq j$ then $x_i \perp x_j | x_{-\{i,j\}}$. From Q we can thereby conclude that if $\phi \neq 0$ then x_t is conditional dependent only on its neighbours x_{t-1} and x_{t+1} for $t = 2, \dots, n-1$. The last and first observation x_1 and x_n only have one neighbour each, x_2 and x_{n-1}, respectively [7, p. 2].

We now create a graph $G = \{V, E\}$ with a node in V for each observation in x and E only containing the edge (i, j) if x_i is conditional dependent on x_j, i.e. $Q_{ij} \neq 0$. Given the structure observed in Q such a graph for an autoregressive process is illustrated in 4.
Since x follows a multivariate normal distribution with mean $\mu = 0$ and covariance matrix $\Sigma = Q^{-1}$ and G is constructed based on Q as described in the section above, we can conclude that x is a GMRF wtr. G.

Since x is a GMRF and follows the described autoregressive process, the conditional correlation between observation x_i and x_j is given by (5)

$$Corr(x_i, x_j | x_{-\{i,j\}}) = -\frac{Q_{ij}}{\sqrt{Q_{ii}Q_{jj}}} = \begin{cases} \frac{\phi}{1+\phi^2} & \text{if } i,j = 2,\dots,n-1, \quad j = i \pm 1. \\ \frac{\phi}{\sqrt{1+\phi^2}} & \text{if } i,j = 1,2, \quad j = i \pm 1 \vee i,j = n-1,n, \quad j = i \pm 1 \\ 0 & \text{else} \end{cases}$$

It is clear from the above expression that the size of ϕ determines the conditional correlation between the observations. When ϕ is larger the conditional correlation is larger. The conditional correlation is further determining the "spatial" correlation, since each observation only is conditional correlated with its neighbours in time. We can thereby conclude that ϕ determines the "spatial" correlation. This is illustrated in figure 3, where it is seen in the top row that when $\phi = 0.9$, the neighbouring values are more similar compared to when $\phi = 0.1$.
In order to estimate the spatial correlation of x we would therefore like to estimate the parameter ϕ. Given some observation of x, we do this by finding the MLE of ϕ. The log-likelihood is found by taking

log to the density of x given in (7):

$$\ell(\phi|x) = \log\left(\frac{1}{(2\pi)^{n/2}}\right) + \frac{1}{2}\log(|Q|) - \frac{1}{2}x'Qx.$$

We then differentiate this wrt. ϕ and set the results equal to zero, which yields the following result:

$$\begin{aligned} 0 &= -\frac{\phi}{1-\phi^2} + b + \phi a \iff \\ 0 &= a\phi^3 + b\phi^2 + (1-a)\phi - b, \end{aligned}$$

where $b = \sum_{i=2}^{n} x_{i-1}x_i$ and $a = -\sum_{i=2}^{n-1} x_i^2$. This is recognized as a cubic equation and we can thereby use know theory to find it roots and determine the MLE of ϕ, as shown in section 6.3 The results of such an estimation is shown in the bottom row of figure 3 for different values of ϕ and an increasing number of observations. It is seen that when the number of observations included in the estimation is increased, the precision of the estimation is improved. It is also seen that when ϕ is larger, the process converges with fewer observations.

2.3.2 p-neighbourhood GMRF

In this section we will explain another example of how the spatial correlation of a GMRF can be estimated by imposing an additional assumption. The following method is what we end up using in order to incorporate spatial information into the standard NMF method and to estimate the spatiality of the estimated weights from the different NMF models.

Given a vector of observations $x = (x_1, \ldots, x_n)^T$ with spatial information, we assume x is a GMRF wrt. the graph G and with mean μ and precision matrix Q. We then define a p-neighbourhood GMRF by imposing an additional assumption in regards to the structure of Q by defining the entries in Q using the p-neighbourhoods of x as:

$$Q_{ij} = \begin{cases} \kappa & \text{if } i = j \\ \kappa\beta & \text{if } i \sim j \\ 0 & \text{else} \end{cases},$$

where $\kappa > 0$ and $|\beta| < 1$.
This structure of Q implies that observation x_i is only conditional dependent on the observations in its p-neighbourhood. This, in turn, results in the graph G having edges based on the observations neighbourhoods as in the example shown in figure 2. We define such a graph as a p-neighbourhood graph.

From this structure of Q it follows from (5) that the conditional correlation becomes

$$Corr(x_i, x_j|x_{-\{i,j\}}) = -\frac{Q_{ij}}{\sqrt{Q_{ii}Q_{jj}}} = \begin{cases} -\frac{\kappa\beta}{\kappa} = -\beta & \text{if } i \sim j \\ 0 & \text{else} \end{cases}.$$

It is seen that $-\beta$ is equal to the conditional correlation between neighbours, and thereby it can be assumed that β measures how spatially correlated the entries of x are.

We are interested in estimating μ, κ and β in order to estimate the spatial correlation of the p-neighbourhood GMRF. $\hat{\mu}$ is given by the vector where all entries are equal to the mean of x. $\hat{\kappa}$ and $\hat{\beta}$ is found by numerically maximizing the likelihood of x wrt. κ and β using gradient descent. Here we start out by finding the log-likelihood of x, which is given by:

$$\ell(\kappa, \beta|x) = \log\left(\frac{1}{(2\pi)^{n/2}}\right) + \frac{1}{2}\log(|Q|) - \frac{1}{2}(x-\mu)'Q(x-\mu).$$

We then differentiate the log-likelihood wrt. κ and β respectively which, as described in section 6.4, yields the following update rules:

$$\kappa^{i+1} = \kappa^i - \gamma_\kappa \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \kappa}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \kappa}\right]\right),$$
$$\beta^{i+1} = \beta^i - \gamma_\beta \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \beta}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \beta}\right]\right).$$

The process of estimating κ and β in a p-neighbourhood GMRF through gradient descent is described in algorithm 3, section 6.4. In figur 5 we show the results of this estimation process when estimating κ and β from $n = 10, \ldots, 500$ simulated observations from a p-neighbourhood GMRF with $p = 3$, $\beta = -0.3$ and $\kappa = 3$.

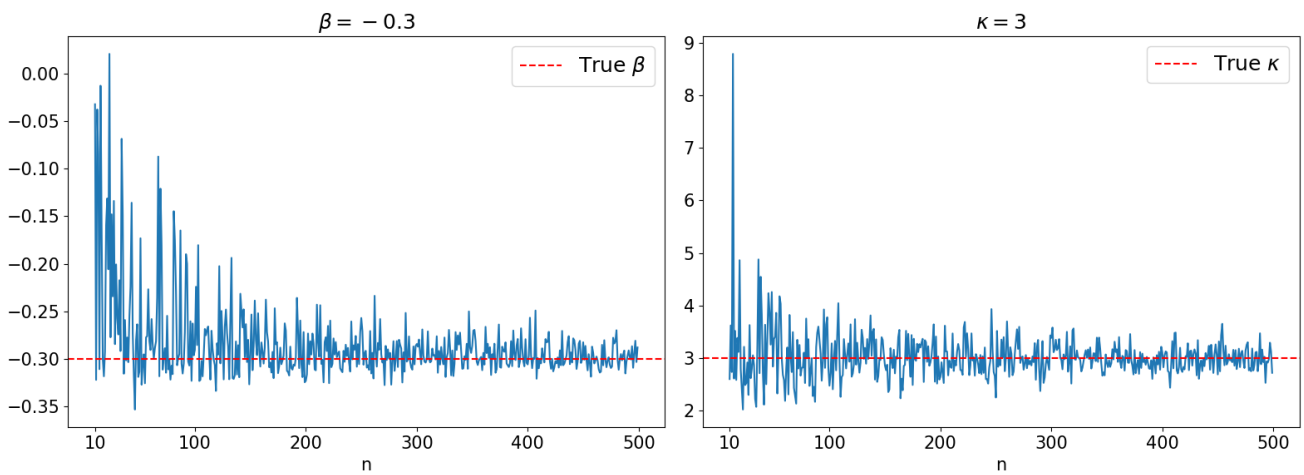

Figure 5: Estimation of κ and β for $n = 10, ..., 500$ simulated observations from a p-neighbourhood GMRF with $p = 3$, $\beta = -0.3$ and $\kappa = 3$.

2.4 GMRF-NMF

In this section we will expand the concept of a spatially aware NMF model from the Simple Spatial NMF, described in section 2.2, to a more advanced model. We do this by including the concept of p-neighbourhood GMRF into the standard NMF model. As the entries in W need to be non-negative, we

assume that each column of W, denoted as $W_{(k)},\ k = 1, \ldots, K$, is a p-neighbourhood log-GMRF wrt. the p-neighbourhood graph, G, based on the spatial information of the cells i.e. a log-normal distribution with the described characteristics of a p-neighbourhood GMRF:

$$\log(W_{(k)}) \sim \text{GMRF}(\mu_k, Q^{(k)}),$$

where $Q_{ij}^{(k)} = \begin{cases} \kappa_k & \text{if } i = j \\ \kappa_k \beta_k & \text{if } i \sim j. \\ 0 & \text{else} \end{cases}$

This model is refereed to as the GMRF-NMF model.

Since the log-GMRF is parameterized through the parameters of its underlying p-neighbourhood GMRF distribution, it is important to note that we are introducing conditional correlations between the entries of $\log(W_{(k)})$ and not $W_{(k)}$ directly. This will give slightly different correlation structure of $W_{(k)}$, but it will be similar to that of $\log(W_{(k)})$ as the log is a monotone transformation.

Analog to the standard NMF, we now assume that:

$$V_{nm}|W \sim \text{Pois}((WH)_{nm}).$$

From this it is possible to define a posterior of W and H, $\pi(W, H|V)$, as defined in [6, section 9.1.1]:

$$\pi(W, H|V) = \frac{\pi(W, H, V)}{\pi(V)} = \frac{\pi(W, H)\pi(V|W, H)}{\pi(V)} \propto \pi(W)\pi(V|W, H).$$

Since H is not stochastic the density, $\pi(W, H)$, reduces to $\pi(W)$, which can be interpreted as the prior distribution of W.

To find new update rules for W and H, we want to maximize the posterior. We defined the loss to be the negative log taken to $\pi(W)\pi(V|W, H)$:

$$\begin{aligned} loss(W, H) &= -\log(\pi(W)) - \log(\pi(V|W, H)) \\ &= \sum_j \left[\frac{1}{2} (\log(W_{(j)}) - \mu_j)^T Q^{(j)} (\log(W_{(j)}) - \mu_j) \right] + D_{GKL}(V||WH) + C, \end{aligned}$$

where C is an additive constant not depending on W and H. Then we differentiated the expression wrt. W and H, respectively, as shown in section 6.5.

This resulted in the following update rule for W:

$$W^{i+1} = W^i - \gamma_W \left(\Lambda^i - \frac{V}{W^i H^i} (H^i)^T + \mathbb{1}^{n \times m} (H^i)^T \right), \tag{8}$$

where $\Lambda^i = \begin{bmatrix} \frac{Q^{(1)}(\log(W^i_{(1)}) - \mu_1)}{W^i_{(1)}} & \cdots & \frac{Q^{(K)}(\log(W^i_{(K)}) - \mu_K)}{W^i_{(K)}} \end{bmatrix}$.

The update rule for H is the same as in the standard model, since only the likelihood $\pi(V|W,H)$ is depending on H:

$$H^{i+1} = H^i - \gamma_H \left(-(W^i)^T \frac{V^i}{W^i H^i} + (W^i)^T \mathbb{1}^{n\times m} \right). \tag{9}$$

For each iteration i in the gradient descent process κ_k and β_k is estimated for $W^i_{(k)},\ k = 1, ..., K$ after W is updated. The full process of estimating W and H through GMRF-NMF is shown in algorithm 4, section 6.5

2.4.1 Simplified GMRF-NMF

In order to avoid estimating κ and β in each iteration, we define a simplified version of GMRF-NMF, where the values of κ_k and β_k are set to be equal: $\kappa_1, \ldots, \kappa_K = \kappa$ and $\beta_1, \ldots, \beta_K = \beta$, and are given as hyperparameters. This results in each $Q^{(k)}$ being equal: $Q^{(1)}, \ldots, Q^{(K)} = Q$. This further results in $\Lambda^i = \frac{Q(\log(W^i)-\mu)}{W^i}$. In the simplified model the update rule for W is:

$$W^{i+1} = W^i - \gamma_W \left(\frac{Q(\log(W^i) - \mu)}{W^i} - \frac{V^i}{W^i H^i}(H^i)^T + \mathbb{1}^{n\times m}(H^i)^T \right),$$

where $\mu = \begin{bmatrix} \mu_1 & \ldots & \mu_K \end{bmatrix}$ and $Q_{ij} = \begin{cases} \kappa & \text{if } i = j \\ \kappa\beta & \text{if } i \sim j \\ 0 & \text{else} \end{cases}$.

The update rule for H is still (9). The full process of estimating W and H through the simplified GMRF is shown in algorithm 5, section 6.5.1.

2.5 Comparing the simple spatial NMF model and the GMRF-NMF model

Since the only difference between the updates of W in the standard NMF (1) and the GMRF-NMF model (8) is Λ added, the update in the GMRF-NMF model can be split into two steps like the update rules for W in the simple spatial model. The two steps are the standard update of W and the weighted p-neighbourhood update. For a specific entry W_{nk} the weighted p-neighbourhood update, after the standard update, becomes:

$$\begin{aligned} W^{i+1}_{nk} &= W^{i+1'}_{nk} + \gamma_W \Lambda^i_{nk} \\ &= W^{i+1'}_{nk} - \gamma_W \left(\frac{\kappa_k(\log(W^i_{nk}) - \mu_k)}{W^i_{nk}} + \sum_{n'\sim n} \frac{\kappa_k\beta_k(\log(W^i_{n'k}) - \mu_k)}{W^i_{n'k}} \right), \end{aligned}$$

where $W^{i+1'}_{nk}$ is the value of W_{nk} at iteration $i+1$ after the standard update.

If we for simplicity assume that all cells have the same amount of neighbours p and $\beta = -\frac{1}{p}$, the update can be expressed as:

$$W_{nk}^{i+1} = W_{nk}^{i+1'} - \gamma_W \kappa_k \left(\frac{(\log(W_{nk}^i) - \mu_k)}{W_{nk}^i} - \frac{1}{\sum_{n'} \mathbb{1}_{n \sim n'}} \sum_{n' \sim n} \frac{(\log(W_{n'k}^i) - \mu_k)}{W_{n'k}^i} \right).$$

Here we see that the parenthesis consists of two terms. The fist term includes information about the specific entry and the last term includes information about the entries neighbours. Both terms compare each entry to the mean of the weight. If the specific entry deviates from the mean then this is captured in the first term which accounts for this when updating. If the neighbours generally deviates from the mean, this is captured in the second term. It is seen that the second term is a weighted average of the neighbours deviations from the mean.

For the simple spatial NMF model, the simple neighbourhood average step is:

$$W_{nk}^{i+1} = \frac{1}{1 + \sum_{n'} \mathbb{1}_{n' \sim n}} \left(W_{nk}^{i+1'} + \sum_{n' \sim n} W_{n'k}^{i+1'} \right).$$

It is seen that the update rule for W in GMRF-NMF is somewhat comparable to the simple spatial NMF. The simple spatial NMF uses a simple p-neighbourhood average after the standard update of W, while the GMRF-NMF subtracts a weighted p-neighbourhood average of the deviations from the mean through Λ after the standard update of W.

3 Results

3.1 Standard NMF VS Simple Spatial NMF

We are interested in evaluating the performance of the standard NMF and the Simple Spatial NMF as described respectively in section 2.1 and 2.2 on the subset of the ST Slide-seqV2 dataset as described i section 1.

Each of the two methods was evaluated on the selected subset with $K = 10$ and Simple Spatial NMF with $p = 5$ for the p-neighbourhoods. After estimating the weights using the two method we assumed that the weights are p-neighbourhood GMRFs with $p = 5$ and estimated κ and β as described in section 2.3.2. For each method the five most spatial weights, i.e. the weights with the most negative β, are shown in figure 6 and 7 alongside the value of β and κ estimated from the respective weight and the gene which are most expressed in the corresponding signature of each weight.

From figure 7 we see that the weights from the Simple Spatial NMF in general are more spatially correlated than those from the Standard NMF (i.e. has a more negative β value) in figure 6 which is expected since we force the weights to be somewhat spatially correlated when using Simple Spatial NMF. Visually the

weights from the Simple Spatial NMF also seems to be more spatially correlated than those from the Standard NMF.

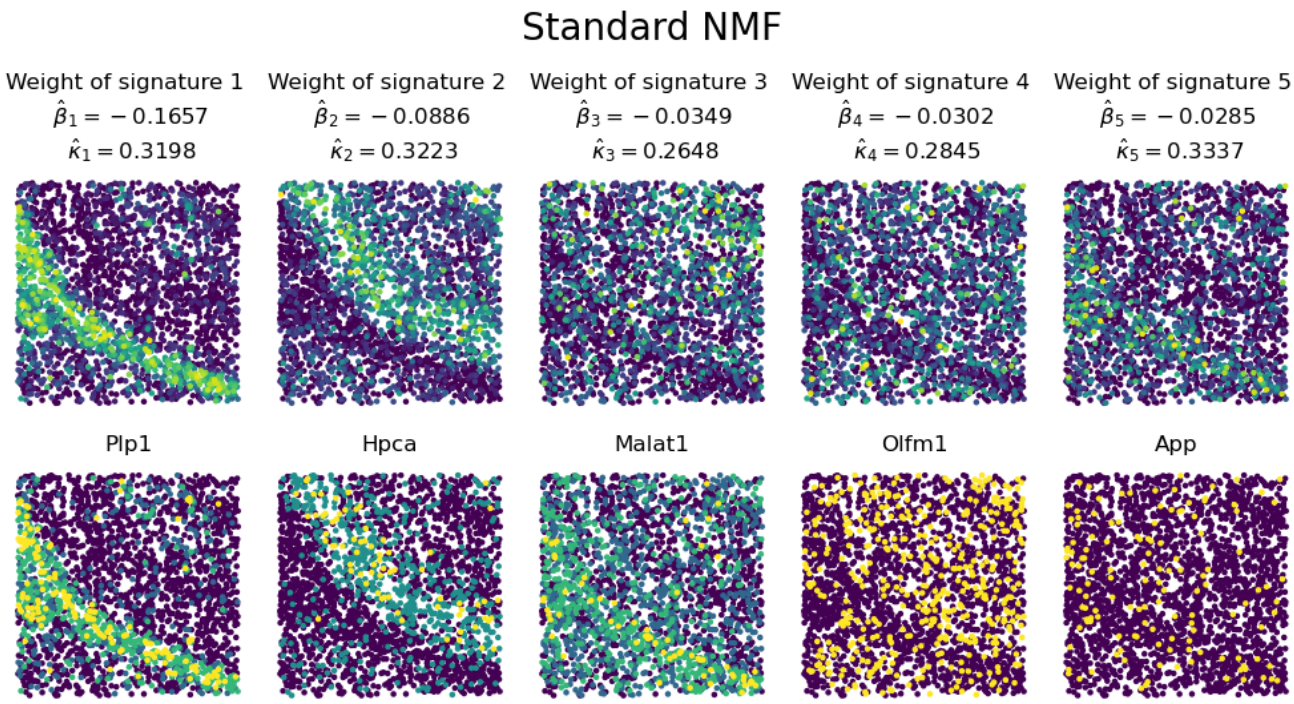

Figure 6: Most spatial weights from the standard NMF model (top row) and the most expressed gene in each of the corresponding signatures (bottom row). For each weight the estimated parameters $\hat{\beta}$ and $\hat{\kappa}$ are shown

We also see that signature 1 and signature 2 in the Standard NMF express genes "Plp1" and "Hpca", resepctively. Both of these genes are also expressed in signature 2 and 3 from the Simple Spatial NMF, respectively. This could indicate that these genes are highly expressed in the data. The weights from the Standard NMF corresponding to these genes, especially the weight of signature 1, which corresponding to "Plp1", have a relatively high estimated spatial correlation. This could further indicate that these genes have a relatively high spatial correlation. A spatial correlation that the Standard NMF is also capable of capturing, likely because of how highly expressed these genes are in the data. We see that the three remaining weights from the Simple Spatial NMF all seem to be somewhat spatially correlated, which is not the case for the weights estimated by the Standard NMF.

As mentioned signature 1 and signature 2 from the standard NMF express the same genes as signature 2 and signature 3 from the Simple Spatial NMF, respectively. This could further indicate that these signatures are pairwise similar, which is emphasized by the fact the corresponding weights are also seen to be visually similar. We, however, see that the corresponding weights from the Simple Spatial NMF has a higher estimated spatial correlation as mentioned above.

All the values of β_k for the weights estimated by the Simple Spatial NMF are around the same value $\beta_k \approx -0.21$. This is expected since the enforced neighbourhood structure is the same for all signatures. The estimated values being close to 1/5, is possibly because a 5-neighbourhood was used in the model, which results in the cells having slightly more than 4 neighbours on average.

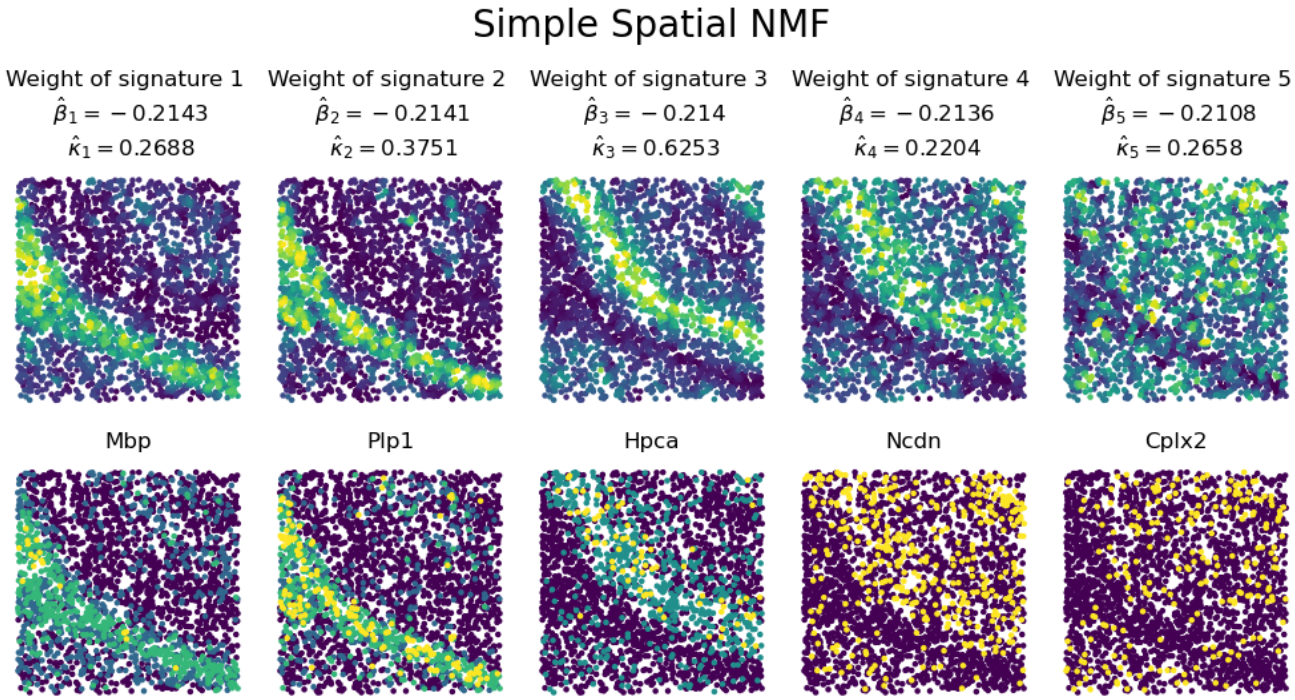

Figure 7: Most spatial signatures from the simple spatial NMF model (top row) and the most expressed gene in each of the corresponding signatures (bottom row). For each weight the estimated parameters $\hat{\beta}$ and $\hat{\kappa}$ are shown

3.2 Results from GMRF-NMF

We further want to evaluate the performance of the GMRF-NMF model, as described in section 2.4, on the same subset of the ST Slide-seq V2 dataset. Here we evaluated both the full method of GMRF-NMF which estimates κ and β for each weight and the simplified one which takes κ and β as hyperparameters and assumes that each weight have the same κ and β value. Both methods were evaluated on the subset with $K = 10$ and $p = 5$ in the p-neighbourhoods. As mentioned the simplified method requires κ and β as hyperparameters. Here these were defined to be the average of the estimated values of κ and β for the different weights in the result produced by the Simple Spatial NMF in an attempt to enforce the same correlation structure. Both models further require multiple learning rates as hyperparameters. Here we used the highest possible learning rate that did not cause the process to fluctuate.

The full version of GMRF-NMF, which estimates κ and β for each of the weights, never converged and therefore did not yield any results for us to evaluate. There is not a clear explanation to why the model did not converge. However a possible reason could be the fact that the model uses a gradient descent process within another gradient descent process and therefore is facing some numerical issues. Another possible reason is that it could have something to do with the assumption of a general κ and β for each weight. The potential reasons and possible solutions will be discussed further in section 4.

The simplified version of GMRF-NMF did manage to converge. However as shown in figure 8 the model did not produce weights and signatures that are able to approximate the original data. Figure 8 shows the residuals for the three different models which have yielded results. It is clear that both the Standard

NMF and the Simple Spatial NMF produces weights and signatures where the corresponding residuals are relatively close to zero with the Standard NMF showing better results compared to the Simple Spatial NMF. This is expected since the Simple Spatial NMF includes further requirements to the weights. Both Standard NMF and Simple Spatial NMF, however, tend to slightly overshoot the zero values and underestimate the higher values in the original data. Both these models are, however, significantly better at approximating the original data compared to the simplified GMRF-NMF model. We see from figure 8 that the simplified GMRF-NMF model tends to severely underestimate the values and in general estimate the values to be between one and zero. The possible reasons for this unsatisfactory result are discussed in section 4.

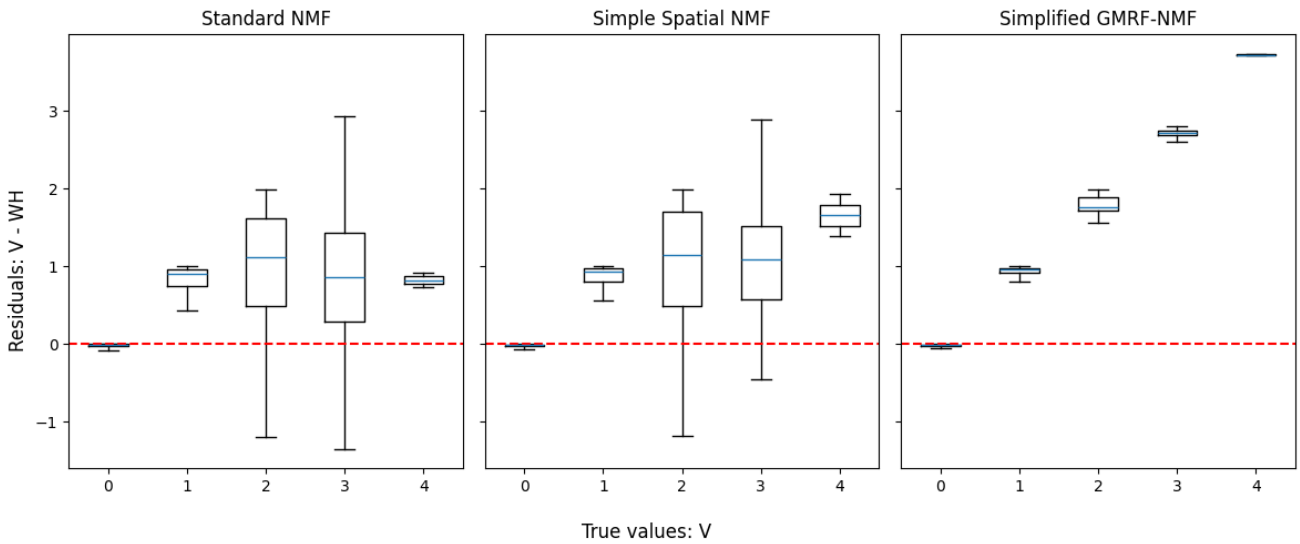

Figure 8: Boxplot of the residuals of the approximation, WH, for the different models, plotted for each corresponding true value of V.

4 Discussion

In the section above, it is seen that both the standard model and the simple spatial model produces interpretable results that generally approximates the original data quite well. The simple spatial model generally captures more spatiality than the standard model which is the desired result. It is further shown that estimating κ and β from the weights, when assuming these are p-neighbourhood GMRFs, generally works well. This is empathized through the fact that the value of β reflects the visual spatiality of the weights and the fact that the weights produced by the simple spatial model generally have more negative values of β than those from the Standard NMF.

4.1 Perspective of Simple Spatial NMF

To include the spatiality of the cells in the weights estimated by the Simple Spatial NMF we use a p-neighbourhood average, as described in section 2.2. This method assumes that each cell is equally correlated with all its p-neighbours. This assumption simplifies the model but is not necessarily realistic.

It seems possible that the correlation between a cell and its neighbours is dependent on the distance to said neighbours. Therefore a more realistic representation of the spatiality between the cells could be incorporated in the simple spatial NMF by weighing the p-neighbourhood average by using a kernel function wrt. the distance between points. It could also be relevant to develop a model without the p-neighbourhoods that instead are based on a kernel average of all points wrt. distance.

4.2 Numerical challenges of GMRF-NMF

Unfortunately, both our simplified and full GMRF-NMF model did not produce results that were interpretable or approximated the original data well. This can be due to different problems that will be discussed in the following.

Both the full and simplified GMRF-NMF model uses gradient descent to estimate the parameters in the model (as described in algorithm 4 and algorithm 5, respectively). Numerical optimization using gradient descent is generally prone to converging to local minima and therefore not yielding a satisfactory result, especially with non-optimal learning rates. Here gradient descent requires setting a learning rate for each of the parameters that we are estimating. The full and simplified model requires respectively four and two learning rates as hyperparameters. The chosen combination of hyperparameters affect the performance of the model significantly. If the learning rates are to high the process might even start to fluctuate instead of converging. A further problem with gradient descent is that it is a computational slow process and therefore takes a lot of time before converging. This, in turn, makes it hard to optimize the combination of learning rates through hyperparameter search since this requires running the process multiple times. The fact that we use a gradient descent process within another gradient descent process in the full GMRF-NMF model just amplifies these issues and might be what are causing the model not to converge. The issue of converging to a local minima might be what are causing the poor performance of the simplified GMRF-NMF model.

One way to deal with the above mentioned issues could be to simply remove the learning rates used for estimating W and H as hyperparameters from the gradient descent processes in the GMRF-NMF models. This is done in the standard NMF and simple spatial NMF by using a specific learning rate that makes the updates *multiplicative.* It might be possible to define a corresponding learning rate in the case of the update rules in the GMRF-NMF models and make these multiplicative. This could potentially improve the generalizability and performance of the models significantly.

Another way to improve the full GMRF-NMF model is to change the way we estimate κ and β in a p-neighbourhood GMRF. Here an obvious improvement would be to find analytical estimates for κ and β, instead of using numerical estimates through gradient descent, as done when estimating ϕ in the autoregressive model. This would remove the issues arising when using gradient descent to estimate κ and β. Another way to potentially improve the estimation process of κ and β in the full GMRF-NMF model is to use Newton-Raphson instead of gradient descent. This would remove the learning rates from the model and possibly improve the runtime.

Both of these improvements in regards to estimating κ and β, however, requires using the entries in the covariance matrix of the corresponding p-neighbourhood GMRF, $\Sigma = Q^{-1}$, directly. When trying to find an analytical estimate for κ and β, the derivatives wrt. κ and β of the likelihood should be set equal to zero and κ and β isolated. This requires looking at the individual entries of Σ, since Σ is depending on κ and β. The case is the same when using Newton-Raphson, since this requires finding the second order derivatives wrt. κ and β, and this again involves the entries of Σ.
It is shown in section 6.3 that, in the case of autoregression, it is possible to define Σ through its entries, Σ_{ij}, however, looking at the entries of Σ of a p-neighbourhood GMRF did not yield a satisfactory way of defining Σ_{ij}. This made it impossible for us to utilize an analytical estimate or the Newton-Raphson method when estimating κ and β.

To completely remove the problem with estimating κ and β in the full GMRF-NMF model, these are fixed to be the same for all weights and given as hyperparameters in the simplified GMRF-NMF. We are however left with other issues regarding this model as described above and as seen in section 3, fixing κ and β did not yield a model which creates meaningful weights and signatures.

4.3 Challenges regarding the parameterization of GMRF-NMF

When fixing κ and β to the same values for each weight, as done in the simplified GMRF-NMF model, we impose a very strict assumption about the correlation of the cells. Here each weight is assumed to have the same degree of spatial correlation no matter which genes the corresponding signature expresses. This assumption might be too strict since we could imagine that some genes are more spatially correlated than others.

When looking at the full GMRF-NMF model, the way κ and β are defined could still also be too strict: The parameters can vary between different weights, which allows the weights to express different degrees of spatiality, but each weight $W_{(k)}$ only has a single value for β_k and for κ_k, which implies that for a given $W_{(k)}$, the correlation between all neighbours is the same.
This might be unrealistic since it is possible to imagine that some cells could be more correlated with its neighbours than other cells are. It is also likely that a cell is more correlated with some of its neighbours than others. If we look at figure 1 of the selected subset we would expect that a cell on the edge of the band is likely more correlated with its neighbours inside the band than its neighbours outside of the band. One of the issues that might arise from using such a structure of κ and β is given by the fact that if the data is very sparse, the sparsity might result in the zeroes being to dominant and thereby making all values of the approximation (WH) too low, since the cells are equally correlated with all its neighbours. This might be what are causing the underestimation of the results from the simplified GMRF-NMF.

A way to deal with this problem would be to let each cell vary its correlation with its neighbours. This could be done by letting κ_k and β_k vary so that for each weight $W_{(k)}$ each cell i has corresponding parameters $\kappa_{k,i}$ and $\beta_{k,ij}$ for each other cell j. That is for each weight each cell has a parameter that controls how correlated the cell is with each of the other cells. If $\beta_{k,ij}$ vary freely, it allows for correlation between

all points with no regard to neighbourhoods. This, however, would make the spatial information of the data redundant, since it is no longer included when updating the parameters. The spatial information could still be including through the p-neighbourhood framework by defining $\beta_{k,ij} = 0$ if i and j are not neighbours.

A significant downside of letting $\kappa_{k,i}$ and $\beta_{k,ij}$ vary is the fact that it would result in a heavily parameterized model. Here we would have up to $k(p+1)$ parameters regarding the correlation for each cell, if the p-neighbourhood framework is included. A further downside would be the fact that the result of such an overparameterized model would not be very interpretable when analysing the general spatiality of the genes in a given signature, since it would yield multiple correlation estimates for the corresponding weight. Here the full GMRF-NMF model, as described in section 2.4, would yield a more interpretable results since it generalizes the spatiality of $W_{(k)}$ corresponding to the signature k in one general κ_k and β_k.

4.4 Complications of using log-GMRF in GMRF-NMF

A further possible issue with the GMRF-NMF models is the general assumption regarding the distribution of the weights. Here we assume that the weights, $W_{(k)}$, are log-GMRFs such that $W_{(k)}$ are ensured to have positive values. This in turn leads to us taking the logarithm to the weight in order to estimate the parameters of that given weight. When we then use the difference between the entries of $W_{(k)}$ and μ_k this is for the log-transformed values of $W_{(k)}$. The log-transformed value of $W_{(k)}$ will have smaller differences than the original values of $W_{(k)}$ and might therefore tone down the spatial information used in the model. A way to fix this possible issue could be to assume directly that $W_{(k)}$ is a p-neighbourhood GMRF and then assume that $V \approx e^W H$ such that we ensure the weight matrix used in approximating V has positive values, as done in [10].

4.5 Hyperparameters

The results from the different methods described in section 3 was created using $K = 10$ signatures in all models and $p = 5$ neighbours in the Simple Spatial and GMRF-NMF models. The different parameters were chosen to be the same value for the different models in order to allow comparison of the results across the models. For the sake of simplicity, these values were chosen arbitrarily.
A different approach for choosing K could have been to find K for each model that would maximize performance. This could for example have been done by using the concept of the Bayesian information criterion (BIC). Here we could have calculated the BIC for each model for different K and chosen the parameter with the lowest BIC. This could however result in different parameter values for the different models which could make it harder to compare the results across the different models. As mentioned before, the runtime of the GMRF-NMF models made it very hard to optimize hyperparameters, and would also make this approach hard to utilize.

5 Conclusion

In this project we have proposed different methods to incorporate spatial information into an NMF model and thereby making the estimated weights more spatial. We have also proposed a way to estimate how spatially correlated a weight is through assuming that the weight is a p-neighbourhood GMRF and estimating the parameters of this GMRF.

The proposed NMF models were evaluated on the ST Slide-seqV2 mouse hippocampus dataset.
As shown in figure 7 the estimated weights from using the Simple Spatial NMF, which utilized a neighbourhood average when estimating the weights, are visually more spatial when compared to the estimated weights from the standard NMF model in figure 6. The weights from the Simple Spatial NMF Model also proved to be more spatially correlated when they are assumed to be a p-neighbourhood GMRF. From this it can be concluded that the Simple Spatial NMF model successfully incorporated some spatial information when estimating the factor matrices.

The two NMF models where the weights were assumed to follow a p-neighbourhood GMRF did not yield any satisfactory results. The full model, where the parameters of the p-neighbourhood GMRF are estimated, did not manage to converge. The simplified model, where all weights are assumed to follow the same p-neighbourhood GMRF and the parameters are given as hyperparameters, did manage to converge, but the results did not prove to be interpretable.
The reasons for this and potential ways to better the models was discussed.

The estimation of the parameters of a p-neighbourhood GMRF was proven to successfully approximate the true parameters of a p-neighbourhood GMRF when used on simulated data. When used on the estimated weights from the NMF models, the estimated parameters generally reflected the expected correlation structure and the visual spatiality of the weights.

Overall, it can be concluded that spatial information was successfully incorporated into an NMF model and the spatiality of the estimated weights were successfully estimated. However, only the Simple Spatial model was successful in incorporating spatial information into the model and producing useful results. The models based on a p-neighbourhood GMRF did not prove to produce satisfactory results.

6 Methods

6.1 Standard NMF

Assuming that $V_{nm} \sim \text{Pois}((WH)_{nm})$ as described in section 2.1. The log-likelihood of V is then given by:

$$\begin{aligned}\ell(W,H|V) &= \underbrace{-\sum_n \sum_m \left[V_{nm} \log\left(\frac{V_{nm}}{WH_{nm}}\right) - V_{nm} + WH_{nm} \right]}_{D_{GKL}(V||WH)} + \underbrace{\sum_n \sum_m \left(V_{nm}\log(V_{nm}) - V_{nm}\right) - \log\left(V_{nm}!\right)}_{C} \\ &= -D_{GKL}(V||WH) + C. \end{aligned} \tag{10}$$

We now want to numerically maximize the log-likelihood wrt. W and H, respectively, using gradient descent.

Since the constant C is only dependent on V, the negative log-likelihood is equal to the generalised Kullback-Leibler up to an additive constant, which then can be used as a loss function when deriving gradient descent update rules for H and W. Firstly, the loss function is rewritten:

$$\begin{aligned} D_{GKL}(V||WH) &= \sum_n \sum_{m'} V_{nm'} \log\left(\frac{V_{nm'}}{(WH)_{nm'}}\right) - V_{nm'} + (WH)_{nm'} \\ &= \sum_n \sum_{m'} V_{nm'} \log\left(\frac{V_{nm'}}{\sum_k W_{nk} H_{km'}}\right) - V_{nm'} + \sum_{k'} W_{nk'} H_{k'm'} \\ &= \sum_n \sum_{m'} V_{nm'} \log\left(V_{nm'}\right) - V_{nm'} \log\left(\sum_{k'} W_{nk'} H_{k'm'}\right) - V_{nm'} + \sum_{k'} W_{nk'} H_{k'm'} \end{aligned} \tag{11}$$

The derivative of the loss function wrt. H_{km} is then given by:

$$\begin{aligned} \frac{\partial D_{GKL}(V||WH)}{\partial H_{km}} &= \sum_n -V_{nm} \frac{1}{\sum_k' W_{nk'} H_{k'm}} W_{nk} + W_{nk} \\ &= -\left[\sum_n W_{nk} \frac{V_{nm}}{(WH)_{nm}} - W_{nk}\right]. \end{aligned} \tag{12}$$

When differentiating the sums $\sum_{k'} W_{nk'} H_{k'm'}$ in (11) only the term including H_{km} derives to W_{nk} while all other terms in these sums derive to 0. This is used when the derivative is found in (12).

The derivative wrt. W_{ia} is found in a similar way and is given by:

$$\frac{\partial D_{GKL}(V||WH)}{\partial W_{nk}} = -\left[\sum_m H_{km} \frac{V_{nm}}{(WH)_{nm}} - H_{km}\right].$$

By using the specific learning rates for H and W the following multiplicative update rules are derived, as shown in 2.1:

$$H_{km}^{i+1} = H_{km}^i \frac{\sum_n W_{nk}^i \frac{V_{nm}}{(W^i H^i)_{nm}}}{\sum_{n'} W_{n'k}^i} = \left(H^i \otimes \frac{(W^i)^T \frac{V}{W^i H^i}}{(W^i)^T \mathbb{1}^{n\times m}}\right)_{km}, \tag{13}$$

$$W_{nk}^{i+1} = W_{nk}^i \frac{\sum_m H_{km}^i \frac{V_{nm}}{(W^i H^i)_{nm}}}{\sum_{m'} H_{km'}} = \left(W^i \otimes \frac{\frac{V}{W^i H^i}(H^i)^T}{\mathbb{1}^{n\times m}(H^i)^T}\right)_{nk}. \tag{14}$$

Algorithm 1: Standard NMF algorithm

Given data matrix V, amount of signatures K and threshold ϵ.

Initialize W^0 and H^0 with random non-negative entries.

for $i = 0, 1, 2, ...$ **do**

  Update signatures H^{i+1} by (13)

  Normalize H^{i+1} so signatures sum to 1

  Update weights W^{i+1} by (14)

  if $D_{GKL}(V||W^iH^i) - D_{GKL}(V||W^{i+1}H^{i+1}) < \epsilon$ **then**

    stop

6.2 Simple Spatial NMF

Since the Simple Spatial NMF uses the same update rules as the standard NMF, and simply further updates W by the p-neighbourhood average, the algorithm becomes:

Algorithm 2: Simple spatial NMF algorithm

Given data matrix V, amount of signatures K, p-neighbourhoods and threshold ϵ.

Initialize W^0 and H^0 with random non-negative entries.

for $i = 0, 1, 2, ...$ **do**

  Update signatures H^{i+1} by (13)

  Normalize H^{i+1} so rows sum to 1

  Update weights W^{i+1} by (14)

  Update all entries in W^{i+1} using a p-neighbourhood average by (3).

  if $D_{GKL}(V||W^iH^i) - D_{GKL}(V||W^{i+1}H^{i+1}) < \epsilon$ **then**

    stop

6.3 Autoregression Parameter Estimation

Assuming $x = (x_1, \ldots, x_n)^T$ is an autoregressive process of order 1 and with normal errors with mean zero and variance $\frac{1}{(1-\phi^2)}$ then the log-likelihood of x becomes

$$\ell(\phi|x) = \log\left(\frac{1}{(2\pi)^{n/2}}\right) + \frac{1}{2}\log(|Q|) - \frac{1}{2}x'Qx.$$

Here we want to find the MLE of ϕ so we find the derivative of the log-likelihood wrt. ϕ

$$\begin{aligned}\frac{\partial \ell(\phi|x)}{\partial \phi} &= \frac{1}{|Q|}|Q|\frac{1}{2}Tr\left[\Sigma\frac{\partial Q}{\partial \phi}\right] + \left(-\frac{1}{2}x'\frac{\partial Q}{\partial \phi}x\right) \\ &= \frac{1}{2}\left(Tr\left[\Sigma\frac{\partial Q}{\partial \phi}\right] - x'\frac{\partial Q}{\partial \phi}x\right).\end{aligned} \tag{15}$$

where

$$\Sigma = Q^{-1}, \qquad \Sigma_{ij} = \frac{1}{1-\phi^2}\phi^{|i-j|}, \qquad Tr\left[\Sigma\frac{\partial Q}{\partial \phi}\right] = \frac{-2\phi}{1-\phi^2},$$
$$x'\frac{\partial Q}{\partial \phi}x = -x_1x_2 + \sum_{i=2}^{n-1} x_i(-x_{i-1} + 2\phi x_i - x_{i+1}) - x_{n-1}x_n.$$

In the above differentiation we used the general rule $\frac{\partial |A|}{\partial y} = |A| Tr\left[A^{-1}\frac{\partial A}{\partial y}\right]$ where A is a matrix and y is a scalar [5, p. 8]

From this follows that (15) can be further reduced:

$$\begin{aligned}\frac{\partial \ell(\phi|x)}{\partial \phi} &= -\frac{\phi}{1-\phi^2} + \frac{1}{2}\left[x_1x_2 - \sum_{i=2}^{n-1} x_i(-x_{i-1} + 2\phi x_i + x_{i+1}) - x_{n-1}x_n\right] \\ &= -\frac{\phi}{1-\phi^2} + \sum_{i=2}^{n} x_{i-1}x_i - \phi\sum_{i=2}^{n-1} x_i^2.\end{aligned}$$

This expression is set equal to zero with $b = \sum_{i=2}^{n} x_{i-1}x_i$ and $a = -\sum_{i=2}^{n-1} x_i^2$:

$$\begin{aligned}0 &= -\frac{\phi}{1-\phi^2} + b + \phi a \\ &= \phi - (1-\phi^2)b - (1-\phi^2)\phi a \\ &= \phi - b + \phi^2 b - \phi a + \phi^3 a \\ &= a\phi^3 + b\phi^2 + (1-a)\phi - b.\end{aligned}$$

This is identified as a cubic equation which is depressed (simplified in a convenient manner) by letting $\phi = y - \frac{b}{3a}$:

$$\begin{aligned}0 &= a\left(y - \frac{b}{3a}\right)^3 + b\left(y - \frac{b}{3a}\right)^2 + (1-a)\left(y - \frac{b}{3a}\right) - b \\ &= -\frac{b^3}{27a^2} + \frac{b^2y}{3a} + ay^3 + \frac{b^3}{9a^2} - 2\frac{b^2y}{3a} - ay + \frac{b}{3} + y - \frac{b}{3a} - b.\end{aligned}$$

and dividing by a:

$$\begin{aligned}0 &= -\frac{b^3}{27a^3} + \frac{b^2y}{3a^2} + y^3 + \frac{b^3}{9a^3} - 2\frac{b^2y}{3a^2} - y + \frac{b}{3a} + \frac{y}{a} - \frac{b}{3a^2} - \frac{b}{a} \\ &= y^3 + \frac{3a(1-a) - b^2}{3a^2}y + \frac{2b^3 - 9ab(1-a) + 27a^2(-b)}{27a^3} \\ &= y^3 + py + q.\end{aligned}$$

where $p = \frac{3a(1-a)-b^2}{3a^2}$ and $q = \frac{2b^3-9ab(1-a)+27a^2(-b)}{27a^3}$.

If the depressed cubic equation found above satisfies $4p^3 + 27q^2 < 0$ then it has three real solutions given by [8, p. 104]:

$$y_k = 2\sqrt{-\frac{p}{3}}\cos\left[\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{\frac{-3}{p}}\right) - \frac{2\pi k}{3}\right] \quad \text{for} \quad k = 1, 2, 3.$$

The solutions for the depressed cubic equation y_1, y_2, y_3 can be translated back to solutions for the original cubic equation ϕ_1, ϕ_2, ϕ_3 by:

$$\phi_k = y_k - \frac{b}{3a},$$

for $k = 1, 2, 3$.

In order to verify that a specific found solution for the cubic equation is in fact a maximum for the log-likelihood we need to check that the second derivative of the log-likelihood is equal to a negative value when evaluated in the found solution. The second derivative of the log-likelihood is given by:

$$\begin{aligned}\frac{\partial \ell(\phi|x)}{\partial^2 \phi} &= \frac{\partial}{\partial \phi} a\phi^3 + b\phi^2 + (1-a)\phi - b \\ &= 3a\phi^2 + 2b\phi + (1-a).\end{aligned}$$

Empirically it was observed that when solving the equation for ϕ the value of two of the roots were -1 and 1, which are not in the range of ϕ. The third root was, however, observed to be in the range of ϕ and proved to be a maximum when evaluating the second derivative of the log-likelihood. Therefore this third root was used as the MLE of ϕ.

6.4 p-neighbourhood GMRF Parameter Estimation

Assuming $x = (x_1, \ldots, x_n)^T$ is a p-neighbourhood GMRF wrt. G with mean μ and precision matrix $Q > 0$ with the structure of Q and G as described in section 2.3.2. Then the log-likelihood of x becomes

$$\ell(\kappa, \beta|x) = \log\left(\frac{1}{(2\pi)^{n/2}}\right) + \frac{1}{2}\log(|Q|) - \frac{1}{2}(x-\mu)'Q(x-\mu). \tag{16}$$

Here we would like to numerically estimate the MLE of κ and β by using gradient descent. Here we define a loss function as the negative log-likelihood of x: $loss(\kappa, \beta) = -\ell(\kappa, \beta|x)$. We then derive the update rules to be used in the gradient descent process by respectively taking the derivative of this loss function wrt. κ and β.

Firstly we find the derivative wrt. κ, as following:

$$\begin{aligned}\frac{\partial loss(\kappa, \beta)}{\partial \kappa} &= -\left(\frac{1}{|Q|}|Q|\frac{1}{2}Tr\left[\Sigma\frac{\partial Q}{\partial \kappa}\right] + \left(-\frac{1}{2}(x-\mu)'\frac{\partial Q}{\partial \kappa}(x-\mu)\right)\right) \\ &= \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \kappa}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \kappa}\right]\right),\end{aligned}$$

where $\left(\frac{\partial Q}{\partial \kappa}\right)_{ii} = 1$ and $\left(\frac{\partial Q}{\partial \kappa}\right)_{ij} = \begin{cases} \beta & \text{if } \sim j \\ 0 & \text{else} \end{cases}$.

In the above differentiation we again used the rule $\frac{\partial |A|}{\partial y} = |A| Tr\left[A^{-1}\frac{\partial A}{\partial y}\right]$ where A is a matrix and y is a scalar [5, p. 8].

Then we find the derivative wrt. β, as following:

$$\begin{aligned}\frac{\partial loss(\kappa,\beta)}{\partial \beta} &= -\left(\frac{1}{|Q|}|Q|\frac{1}{2}Tr\left[\Sigma\frac{\partial Q}{\partial \beta}\right] + \left(-\frac{1}{2}(x-\mu)'\frac{\partial Q}{\partial \beta}(x-\mu)\right)\right) \\ &= \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \beta}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \beta}\right]\right),\end{aligned}$$

where $\left(\frac{\partial Q}{\partial \beta}\right)_{ij} = \begin{cases} \kappa & \text{if } i \sim j \\ 0 & \text{else} \end{cases}$.

Here the same rule mentioned above regarding the derivative of a matrix wrt. a scalar was used.

This yields the following update rules

$$\kappa^{i+1} = \kappa^i - \gamma_\kappa \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \kappa}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \kappa}\right]\right), \tag{17}$$

$$\beta^{i+1} = \beta^i - \gamma_\beta \frac{1}{2}\left((x-\mu)'\frac{\partial Q}{\partial \beta}(x-\mu) - Tr\left[\Sigma\frac{\partial Q}{\partial \beta}\right]\right). \tag{18}$$

Using these update rules in the gradient decent process gives us the following algorithm:

Algorithm 3: Gradient descent algorithm for estimating parameters in a p-neighbourhood GMRF

Given observations x from a p-neighbourhood GMRF wrt. p-neighbourhood graph G, learning rate for κ γ_κ, learning rate for β γ_β and threshold ϵ

Initialise κ^0 and β^0 randomly with $\kappa > 0$

for $i = 0, 1, 2, \ldots$ **do**

- Define precision matrix Q for x based on G and κ^i and β^i
- Update κ^{i+1} by (17)
- Update β^{i+1} by (18)
- **if** $loss(\kappa_i, \beta_i) - loss(\kappa_{i+1}, \beta_{i+1}) < \epsilon$ **then**
  - **stop**

6.5 GMRF-NMF Update Rules

Assuming that $W_{(k)}, \quad k = 1, ..., K$ is a p-neighbourhood log-GMRF and $V_{nm}|W \sim \text{Pois}((WH)_{nm})$ as described in section 2.4, the posterior is then proportional to:

$$\pi(W, H|V) \propto \pi(W)\pi(V|W, H).$$

Since $\text{argmax}_W \, \pi(W) = \text{argmax}_W \, \pi(\log(W))$, the posterior can be maximized, by maximizing:

$$\pi(\log(W))\pi(V|W, H). \tag{19}$$

To define a loss function, the negative log is taken to 19:

$$loss(W, H) = -\log(\pi(\log(W))\pi(V|W, H)) = -\log(\pi(\log(W)) - \log(\pi(V|W, H)). \tag{20}$$

The first term is simply a sum over negative log-likelihoods of GMRFs as in (16) since the columns of W are assumed to be p-neighbourhood log-GMRFs and independent. The second term is simply the negative log-likelihood from the standard model as in (10).
The loss in (20) is then given by:

$$\begin{aligned} loss(W, H) = &-\sum_j \left[\log((2\pi)^{-n/2}) + \frac{1}{2}\log(|Q^{(j)}|) - \frac{1}{2}(\log(W_{(j)}) - \mu_j)^T Q^{(j)}(\log(W_{(j)}) - \mu_j)\right] \\ &+ \underbrace{\sum_n \sum_m \left[V_{nm} \log\left(\frac{V_{nm}}{WH_{nm}}\right) - V_{nm} + WH_{nm}\right]}_{D_{GKL}(V||WH)} - C. \end{aligned} \tag{21}$$

where $C = \sum_n \sum_m (V_{nm} \log(V_{nm}) - V_{nm}) - \log(V_{nm}!)$.

To derive an update rule for W, the derivative of (21) wrt. W_{ia} is found. For simplicity the additive constants are removed from (21) before differentiating:

$$\begin{aligned} \frac{\partial loss(W, H)}{\partial W_{nk}} &= \frac{\partial}{\partial W_{nk}} \sum_j \left[\frac{1}{2}(\log(W_{(j)}) - \mu_j)^T Q^{(j)}(\log(W_{(j)}) - \mu_j)\right] + D_{GKL}(V||WH) \\ &= \left(\frac{Q^{(k)}(\log(W_{(k)}) - \mu_k)}{W_{(k)}}\right)_i - \sum_m \left[H_{km}\frac{V_{nm}}{(WH)_{nm}} - H_{km}\right] \\ &= \left(\Lambda - \frac{V}{WH}H^T + \mathbb{1}^{n\times m}H^T\right)_{nk}, \end{aligned}$$

where $\Lambda = \left[\frac{Q^{(1)}(\log(W_{(1)})-\mu_1)}{W_{(1)}} \quad \dots \quad \frac{Q^{(K)}(\log(W_{(K)})-\mu_K)}{W_{(K)}}\right]$.

The gradient descent update rule for W then becomes:

$$W^{i+1} = W^i - \gamma_W \left(\Lambda^i - \frac{V}{W^i H^i} (H^i)^T + \mathbb{1}^{n \times m} (H^i)^T \right). \tag{22}$$

The update rule for H is the same as the non-multiplicative update rule for the standard NMF:

$$H^{i+1} = H^i - \gamma_H \left(-(W^i)^T \frac{V^i}{W^i H^i} + (W^i)^T \mathbb{1}^{n \times m} \right). \tag{23}$$

This gives the following algorithm:

Algorithm 4: GMRF-NMF algorithm

Given data matrix V, p-neighbourhood graph G, amount of signatures K, learning rates $\gamma_H, \gamma_W, \gamma_\kappa, \gamma_\beta$ and threshold ϵ.
Initialize W^0 and H^0 with random non-negative entries.
Initialise κ_j^0 and β_j^0 for $j = 1, 2, ...K$ randomly with $\kappa_j > 0$
for $i = 0, 1, 2, ...$ **do**
  Update signatures H^{i+1} by (23)
  Normalize H^{i+1} so signatures sum to 1
  Update weights W^{i+1} by (22)
  for $j = 1, 2, \ldots K$ **do**
    Estimate κ_j^{i+1} and β_j^{i+1} for $W_{(j)}^{i+1}$ by algorithm 3 with κ and β initialized as κ_j^i and β_j^i.
  if $loss(W^i, H^i) - loss(W^{i+1}, H^{i+1}) < \epsilon$ **then**
    stop

6.5.1 Simplified GMRF-NMF model

As shown in section 2.4.1 a simplified model, where κ and β are given as hyperparameters is defined. W is then updated as:

$$W^{i+1} = W^i - \gamma_W \left(\frac{Q(\log(W^i) - \mu}{W^i} - \frac{V^i}{W^i H^i} (H^i)^T + \mathbb{1}^{n \times m} (H^i)^T \right), \tag{24}$$

where $Q_{ij} = \begin{cases} \kappa & \text{if } i = j \\ \kappa\beta & \text{if } i \sim j \\ 0 & \text{else} \end{cases}$ and $\mu = \begin{bmatrix} \mu_1 & \ldots & \mu_K \end{bmatrix}$.

This yields the following algorithm:

Algorithm 5: Simplified GMRF-NMF algorithm

Given data matrix V, p-neighbourhood graph G, amount of signatures K, learning rates γ_H, γ_W, p-neighbourhood GMRF parameters κ and β and threshold ϵ.

Initialize W^0 and H^0 with random non-negative entries.

for $i = 0, 1, 2, ...$ **do**

  Update signatures H^{i+1} by (23)

  Normalize H^{i+1} so signatures sum to 1

  Update weights W^{i+1} by (24)

  if $loss(W^i H^i) - loss(W^{i+1} H^{i+1}) < \epsilon$ **then**

    stop

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